

Lecture 7: Strong Induction

Matches Game

- Two piles of equal numbers of matches.
- Player 1 picks a match from a pile of his or her choosing.
- Player 2 picks a match from a pile of his or her choosing.
- A player loses when he or she cannot pick a match.

Base Case (simple game)

- $n = 0$
- Two piles of 0 matches.
- Player 1 loses immediately.

Inductive Case (simple game)

- $n = k+1$
- Player 1 picks from pile i .
- Now pile i has k matches.
- Let player 2 pick from the other pile.
- Now each pile has k matches.
- Inductive hypothesis: In a k pile game, Player 2 can always win.

Better Matches Game

- Two piles of equal numbers of matches.
- Player 1 picks *any number of* matches from a pile of his or her choosing.
- Player 2 picks *any number of* matches from a pile of his or her choosing.
- A player loses when he or she cannot pick a match.

Base Case (better game)

- $n = 0$
- Two piles of 0 matches.
- Player 1 loses immediately.

Inductive case

- IH: In a k matches game, player 2 can always win.
- Consider the $k+1$ matches game.
- Suppose player 1 picks 4 matches from the first pile.
- ???

Strong Inductive case

- IH: In any game with less than $k+1$ matches, player 2 can always win.
- Consider the $k+1$ matches game.
- Suppose player 1 picks m matches from the pile i .
- Let player 2 pick m matches from pile j (the other pile).
- Now we're in a $k+1-m$ pile game.
- So by our hypothesis player 2 can always win.

Strong Induction

- If P is true in the base case (typically means $n = 0$)
- P being true in all cases less than k implies P is true for k .
- Then P is true for all numbers.

Factorization

- Claim: Every number greater than 1 is the product of prime numbers.

Base Case

- $n = 2$.
- 2 can be expressed as 2.

Inductive case

- SIH: For all $x < k$, $1 < x \rightarrow x$ can be expressed as the product of primes.
- What about k ?
- Suppose k is prime: $k = k$ and we're done.
- Suppose k is composite.
 - There's $x < k$ and $y < k$, $x * y = k$.
 - x can be expressed as the product of primes ($x_1 * \dots * x_n$) and so can y ($y_1 * \dots * y_n$).
 - $k = x_1 * \dots * x_n * y_1 * \dots * y_n$